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Executive Summary

This project examines the problem of market “regime”-based portfolio optimization using Hidden Markov Models (HMM). Financial markets are known to exhibit time-varying behavior, with periods of low volatility and steady growth interrupted by episodes of heightened uncertainty and market stress. Traditional portfolio optimization techniques, such as mean-variance optimization, typically assume stable return distributions over time and may therefore fail to adapt to these changing regimes. In this study, market regimes are defined as latent states characterized primarily by differences in return volatility and mean behavior. The objective is to determine whether incorporating regime information can improve portfolio performance relative to traditional static allocation strategies.

To achieve this, Hidden Markov Models are used to identify latent market regimes based on historical return data. A diversified basket of eight ETFs is selected to construct portfolios, with the potential for extension to broader asset classes in the future. Both two-state and three-state HMM specifications are considered. The two-state model captures low- and high-volatility environments, while the three-state model introduces an intermediate regime representing transitional market conditions. For each identified regime, optimal portfolios are constructed using mean-variance optimization based on regime-specific expected return and covariance estimates.

Two portfolio construction approaches are implemented. First, a hard regime-switching strategy allocates fully to the portfolio corresponding to the most likely regime. Second, a probability-weighted approach constructs portfolios as weighted combinations of regime-specific allocations, using the posterior probabilities from the HMM. This probabilistic approach allows for smoother transitions between regimes and reduces sensitivity to regime misclassification.

The models are trained using daily ETF return data from 2007 to 2018 and evaluated through out-of-sample backtesting over the period 2019 to 2025, ensuring no look-ahead bias through the use of lagged signals. Portfolio performance is compared against benchmark strategies, including equal-weight and static mean-variance portfolios. The results demonstrate that regime-aware strategies consistently outperform traditional benchmarks. The probability-weighted two-state model achieves an annualized return of approximately 19.6% with a Sharpe ratio of 1.59, compared to 0.80 and 0.67 for the equal-weight and static portfolios, respectively. The hard regime-switching two-state model achieves a comparable annualized return of approximately 19.3%, but at a higher volatility cost, resulting in a lower Sharpe ratio of approximately 1.46.

Further analysis shows that extending the framework to a three-state model improves risk-adjusted performance by allowing for more gradual adjustments in portfolio allocations at the cost of lower absolute return performance. The probability-weighted three-state model achieved a Sharpe ratio of approximately 1.61 with lower volatility, but reduced annualized returns of 16.4%. Similar to the results of the two-state framework, the three-state hard regime-switching strategy underperforms, achieving an annualized return of approximately 14.6% and a Sharpe ratio of 1.34. Overall, the results suggest that incorporating regime dynamics and probabilistic portfolio adjustments can lead to substantial improvements in both absolute and risk-adjusted performance, while also enhancing robustness across different market conditions.

1. Introduction

Financial markets are inherently dynamic, exhibiting periods of stability and growth interspersed with episodes of heightened volatility and market stress. These shifts in behavior are often driven by changes in macroeconomic conditions, investor sentiment, and external shocks such as financial crises or geopolitical events. As a result, asset return distributions are not constant over time, and the assumption of stationarity underlying many traditional portfolio optimization techniques may not hold in practice.

One of the most widely used approaches in portfolio construction is mean-variance optimization, which relies on estimates of expected returns and covariance matrices. While effective under stable conditions, this framework assumes that these parameters remain constant over time. In reality, empirical evidence suggests that both volatility and correlation structures vary significantly across different market environments. Consequently, a static portfolio allocation strategy may fail to adapt to changing conditions, potentially leading to suboptimal performance and increased exposure to risk during adverse periods. The foundational work of Markowitz [1] provides the basis for this approach, but does not explicitly account for time-varying market dynamics.

This project examines the problem of incorporating time-varying market behavior into portfolio construction using regime-switching models. Specifically, Hidden Markov Models (HMMs) are used to identify latent market regimes from historical return data. These regimes are defined as unobservable states characterized by distinct statistical properties, particularly differences in return volatility and mean behavior. HMMs have been widely used for modeling time series with hidden states, as described in the work of Rabiner [2], and have been applied to finance to capture regime changes in economic and financial data [3].

Two modeling frameworks are implemented: a two-state model capturing low- and high-volatility regimes, and a three-state model that additionally includes an intermediate medium-volatility regime representing transitional market conditions. For each identified regime, optimal portfolio weights are determined using mean-variance optimization based on regime-specific estimates of expected returns and covariance matrices. In each modeling framework, two portfolio construction strategies are used: a hard regime-switching strategy, where portfolios are adjusted discretely based on the most likely regime, and a probability-weighted strategy, where portfolios are constructed as weighted combinations of regime-specific allocations using posterior probabilities from the HMM. The probabilistic approach is intended to create smoother transitions between regimes and reduce sensitivity to regime misclassifications.

The models are evaluated using an out-of-sample backtesting framework designed to avoid look-ahead bias. Daily return data for a diversified basket of exchange-traded funds (ETFs) is used to construct and test the portfolios, sourced from publicly available financial databases [4]. Portfolio performance is compared against benchmark strategies, including equal-weight and static mean-variance portfolios. The results demonstrate that regime-aware strategies significantly outperform traditional benchmarks. In particular, probability-weighted approaches outperform hard-switching strategies on an annualized return and volatility basis.

Overall, the results suggest that incorporating regime dynamics and probabilistic portfolio adjustments can lead to substantial improvements in both absolute and risk-adjusted performance, while also enhancing robustness across different market conditions.

2. Problem Statement

Traditional portfolio optimization methods, such as mean-variance optimization, rely on the assumption that asset return distributions are stable over time. In practice, however, financial markets exhibit significant time variation in both volatility and return characteristics. Periods of low volatility and steady growth are often interrupted by episodes of heightened uncertainty in which correlations increase, and asset behavior changes substantially. As a result, portfolios constructed under the assumption of constant parameters may become suboptimal when market conditions shift.

The primary problem addressed by this project is whether portfolio performance can be improved by explicitly accounting for changes in market conditions through the identification and use of market regimes. In particular, the study investigates whether conditioning portfolio allocation on latent market regimes–defined by differences in return volatility and mean behavior–can lead to superior performance compared to traditional static allocation strategies.

To address this problem, it is necessary to identify distinct market regimes using observable sequences from financial data. Since regimes themselves are not directly observable, a statistical framework is required to infer them from historical returns. Hidden Markov Models provide a natural approach for this problem by modeling the market as a sequence of latent states with different statistical properties. The challenge then becomes how to effectively incorporate the inferred regime information into portfolio construction in a way that improves both return and risk characteristics.

This project focuses on constructing regime-dependent portfolios using a diversified set of exchange-traded funds. For each identified regime, optimal portfolio weights are computed using mean-variance optimization based on regime-specific estimates of expected returns and covariance matrices. Two portfolio construction strategies are considered: a hard regime-switching strategy, which fully reallocates the portfolio based on the most likely regime, and a probability-weighted strategy, which adjusts portfolio weights continuously based on the posterior probabilities derived from the HMMs.

The central hypothesis of this project is that portfolio strategies conditioned on inferred market regimes will outperform traditional static allocation strategies in both absolute and risk-adjusted performance metrics. This hypothesis is tested through out-of-sample backtesting using metrics such as annualized return, volatility, Sharpe ratio, and maximum drawdown. By comparing regime-based strategies to benchmark portfolios, the study aims to quantify the incremental value of incorporating regime information into portfolio construction.

The effectiveness of these approaches is evaluated using out-of-sample data and lagged signals to ensure that results are not driven by look-ahead bias and overfitting. The objective of this project is therefore to determine whether incorporating regime information into portfolio optimization leads to measurable

improvements in performance and robustness, while maintaining realistic and implementable portfolio strategies.

3. Approach

This section describes the methodology used to identify market regimes and construct regime-dependent portfolios. The approach consists of three main components: (1) identification of latent market regimes using Hidden Markov Models, (2) construction of optimal portfolios for each regime using mean-variance optimization, and (3) evaluation of the resulting strategies through out-of-sample backtesting. The objective is to assess whether incorporating regime information leads to improved portfolio performance relative to traditional benchmark strategies.

3.1 Data

The analysis is conducted using daily return data for a diversified set of exchange-traded funds (ETFs), representing multiple asset classes. The selected ETFs incorporate exposure to equities, fixed income, and alternative assets, which provides a broad representation of market behavior. The assets used in this study include: SPY (U.S equities), QQQ (technology equities), IWM (small-cap equities), EFA (international equities), LQD (investment-grade corporate bonds), HYG (high-yield bonds), TLT (long-term U.S. Treasury bonds), and GLD (gold).

The use of ETFs instead of individual equities is a deliberate modeling choice. ETFs provide broad exposure to specific asset classes or market segments, allowing the analysis to focus on systematic market behavior rather than idiosyncratic firm-level risk. Individual equities are often subject to company-specific events, such as earnings announcements, management changes, or sector-specific shocks, which introduce noise unrelated to overall market conditions. Since the objective of this study is to identify and model market regimes, it is desirable to use assets that reflect aggregate market dynamics rather than isolated firm behavior. Furthermore, ETFs offer built-in diversification, which reduces the estimation noise in both expected returns and covariance matrices. This is particularly important in the context of regime-dependent portfolio optimization, as parameters are estimated separately for each regime using subsets of the data. Using individual equities would require a significantly larger cross-section of assets to achieve stable estimates and would increase the dimensionality and complexity of the optimization problem.

From a practical point of view, ETFs also offer advantages, including data availability, liquidity, and consistency. Historical ETF data are readily accessible, with relatively few missing observations, and represent investable portfolios that can be implemented in practice. This makes the results of the analysis more realistic and directly applicable to real-world portfolio management.

Historical price data for these ETFs are obtained from publicly available sources, specifically Yahoo Finance [4]. Daily log returns are computed from adjusted closing prices to ensure consistency and to account for dividends and corporate actions. Log returns are used due to their additive properties and suitability for time series modeling. The dataset is divided into two distinct periods: an in-sample period

from January 2007 to December 2018, and an out-of-sample period from January 2019 to December 2025.

The choice of the 2007 to 2025 time period is motivated by both data availability and the desire to capture a diverse set of market conditions that is also representative of modern market environments. While earlier historical periods, such as the dot-com bubble and subsequent crash of the early 2000s, also represent important market regimes, they are not included in this analysis. This exclusion is primarily due to data consistency constraints and the structure of the selected asset universe. Many of the ETFs used in this study either did not exist or lacked sufficiently liquid and reliable trading histories before the mid-2000s. Including earlier periods would therefore require either reducing the set of assets or introducing missing data, both of which would compromise the stability and comparability of the results. Furthermore, ETFs are becoming increasingly prevalent investment instruments, which may reflect changing investor methodologies and preferences. Consequently, periods without ETFs as a ubiquitous investment vehicle may no longer be representative of modern market environments.

The exclusion of earlier periods does not materially limit the ability of the model to capture market regime behavior. The chosen period from 2007 onward already includes multiple distinct market regimes, including the Global Financial Crisis (GFC) and the COVID-19 market shock, both of which provide clear examples of high-volatility environments. These events offer sufficient variation for the model to learn and identify regime dynamics without requiring additional historical extensions. The in-sample period (2007-2018) is used to estimate model parameters and construct regime-specific portfolios, while the out-of-sample period (2019-2025) is used exclusively for backtesting and performance evaluation. This separation helps to ensure the results are not influenced by look-ahead bias and provides a realistic assessment of the model's predictive capabilities.

In order to identify market regimes, the Hidden Markov Model is trained using the return series of a representative market index. In this case, SPY is used as the market index due to its broad coverage of the U.S. equity market and its ability to capture overall market conditions. The inferred regimes are then applied to the full set of ETFs for portfolio construction and evaluation.

3.2 Hidden Markov Model

To identify market regimes, this study uses a Hidden Markov Model, a statistical model designed to represent systems that evolve over time according to unobservable (hidden) states. In the context of financial markets, these hidden states correspond to different market regimes, such as low-volatility, medium-volatility, and high-volatility environments. The observed data consist of asset returns, while the underlying regime generating these returns is assumed to be latent.

An HMM is characterized by two key components: a hidden state process and an observation process. The hidden state sequence $\{S_t\}$ follows a Markov chain, meaning that the probability of transitioning to a future state depends only on the current state. This is expressed as:

$$P(S_t = j | S_{t-1} = i, S_{t-2}, \ldots) = P(S_t = j | S_{t-1} = i)$$

where $S_t \in \{1, 2, \ldots, K\}$ represents a regime at time t, and K is the number of possible regimes.

The transition dynamics between regimes follow a transition probability matrix:

$$A = \left[a_{ij}\right],\ a_{ij} = P(S_t = j | S_{t-1} = i)$$

where each row of A sums to one. This matrix captures the persistence of regimes and the likelihood of transitions between them.

Conditional on the hidden state, the observed return r_t is assumed to follow a probability distribution specific to that regime. In this study, returns are modeled using a Gaussian distribution:

$$r_t \mid S_t = k \ \sim \ N(\mu_k, \sigma_k^2)$$

where μ_k and σ_k^2 denote the mean and variance of returns in regime k. This formulation allows for each regime to be characterized by distinct statistical properties, particularly differences in volatility and average return.

The parameters of the HMM, including transition probabilities and regime-specific distribution parameters, are estimated using maximum likelihood methods. In practice, this is implemented using the Expectation-Maximization (EM) algorithm, which iteratively updates the parameter estimates to maximize the likelihood of the observed data. The Baum-Welch algorithm, a specific application of EM for HMMs, is used to perform this estimation.

Once the model is fitted, the most likely sequence of hidden states can be inferred using the Viterbi algorithm, and the probability of being in each state at a given time can be obtained using forward-backward procedures. These methods allow the model to assign both a most likely regime and a probability distribution over regimes at each point in time. The former is used in the hard regime-switching strategy, while the latter is used to construct probability-weighted portfolios.

To ensure consistency in interpretation, the inferred states are relabeled based on their estimated volatility. Specifically, regimes are ordered from lowest to highest volatility, allowing for a consistent mapping between model states and economic interpretation. In the two-state model, the regimes correspond to low- and high-volatility environments. In the three-state model, an additional intermediate regime is identified, which represents transitional market conditions.

The HMM is trained using in-sample data, specifically the return series of a representative market index (SPY), which serves as a proxy for overall market conditions. The fitted model is then applied to out-of-sample data to infer regime probabilities without re-estimating parameters, which ensures the regime classification process does not incorporate future information. This approach preserves the integrity of the backtesting framework and avoids look-ahead bias.

Overall, the Hidden Markov Model provides a flexible and statistically grounded framework for identifying market regimes and capturing time-varying behavior in financial returns. By linking these regimes to portfolio allocation decisions, the model enables a dynamic investment strategy that adapts to changing market conditions.

3.3 Regime Identification

Following the estimation of the Hidden Markov Model, the next step is to interpret and label the inferred hidden states as economically meaningful market regimes. Since the states produced by the HMM are unlabeled and arbitrary, a post-estimation procedure is required to assign a consistent interpretation to each state.

To achieve this, the inferred states are ordered based on their estimated volatility. Specifically, the sample standard deviation of returns is computed for observations assigned to each state in the in-sample period. The states are then ranked from lowest to highest volatility, allowing them to be consistently labeled as low-volatility, medium-volatility (in the three-state case), and high-volatility regimes. This ordering ensures that the economic interpretation of each regime remains stable across both in-sample and out-of-sample periods.

For the two-state model, the regimes correspond to a low-volatility regime and a high-volatility regime. The low-volatility regime is characterized by relatively stable returns and lower variance, often associated with periods of market expansion or steady growth. In contrast, the high-volatility regime exhibits significantly higher variance and typically lower or negative average returns, corresponding to periods of market stress or uncertainty. The three-state model introduces an additional intermediate regime that captures transitional market conditions. This medium-volatility regime is characterized by moderate levels of risk and relatively flat or slightly negative returns. Economically, it can be interpreted as a period of increased uncertainty or regime transition, during which the market is neither fully stable nor in a state of crisis. The inclusion of this intermediate regime allows for a more nuanced representation of market dynamics.

Empirical results from the in-sample estimate confirm this interpretation. The low-volatility regime is associated with the highest average returns and lowest volatility, while the high-volatility regime exhibits the lowest average returns and highest volatility. The intermediate regime lies between these two extremes, both in terms of return and risk.

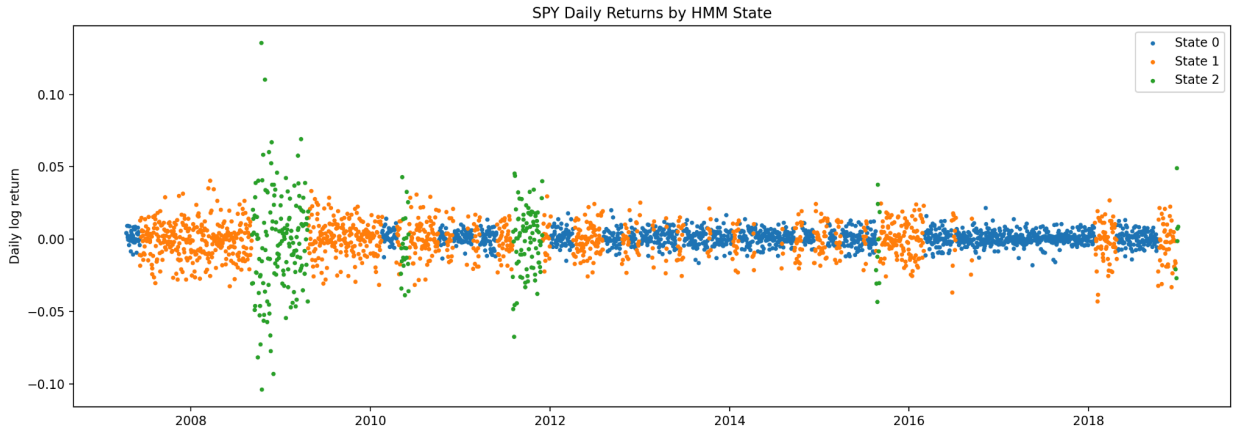

Figure 1: Daily SPY returns colored by inferred market regime (in-sample)

To further illustrate the distinction between regimes, Figure 1 presents a scatter plot of daily returns colored according to the inferred regime classification. The states are labeled as State 0 (blue), State 1 (orange), and State 2 (green), which correspond to the low-, medium-, and high-volatility regimes, respectively. As shown in the figure, the high-volatility regime exhibits the highest variability of returns and corresponds with periods of historical market stress. The low- and medium-volatility states also show distinct behavior associated with their respective means and standard deviations. This visual separation supports the interpretation that the HMM successfully identifies distinct market environments based on return behavior

Additionally, the estimated transition matrix indicates that regimes are persistent, with high probabilities of remaining in the same state. Transitions tend to occur gradually, particularly between low- and medium-volatility states, while shifts into the high-volatility regime are less frequent, but typically correspond to periods of market stress. This behavior is consistent with well-documented features of financial markets.

In the out-of-sample period, the fitted HMM is used to infer regime membership without re-estimating model parameters. This is done by applying the trained model to new data and computing the posterior probability of each regime at each point in time. These probabilities provide a probabilistic measure of regime membership and are used directly in the portfolio construction process. Two types of regime signals are derived from the model. First, the most likely regime at each point in time is obtained by selecting the state with the highest posterior probability. This discrete signal is used in the hard regime-switching strategy. Second, the full vector of posterior probabilities is retained and used to construct probability-weighted portfolios, allowing for smoother transitions between regimes and reducing sensitivity to classification uncertainty.

To avoid look-ahead bias, all regime signals used in portfolio construction are lagged by one period. This ensures that only information available at time $t - 1$ is used to determine portfolio allocations at time t. This step is essential for maintaining a realistic backtesting framework and ensuring that performance results are not artificially inflated. Overall, the regime identification process transforms the statistical output of the HMM into economically meaningful signals that can be directly used for portfolio allocation. By distinguishing between different market environments and assigning probabilities to each regime, the model provides a flexible and robust foundation for dynamic portfolio strategies.

3.4 Portfolio Optimization

Once market regimes have been identified, the next step is to construct portfolio allocations that are tailored to each regime. The objective is to exploit differences in return and risk characteristics across regimes by adjusting asset weights accordingly. This is achieved using a mean-variance optimization framework applied separately within each regime.
For each regime k, expected returns μ_k and covariance matrices Σ_k are estimated using the subset of in-sample data corresponding to that regime. These estimates are then used to construct an optimal portfolio by solving the following optimization problem:

$$\max_{w} \; w^T \mu_k - \lambda w^T \Sigma_k w$$

subject to:

$$\sum_{i=1}^{N} w_i = 1,\ w_i \geq 0$$

where w represents the vector of portfolio weights, λ is a risk-aversion parameter, and N is the number of assets. The constraint $\Sigma w_i = 1$ ensures full investment, while the non-negativity constraint prohibits short-selling. In practice, an additional upper bound is imposed on individual asset weights to prevent excessive concentration in a single asset. In this study, 0.5 is the upper bound imposed to prevent more than 50% concentration in any one ETF.

Solving this optimization problem for each regime yields a set of regime-specific portfolios $\{w_1, w_2, ..., w_K\}$, where each portfolio is optimized for the statistical properties of its corresponding market environment. For example, low-volatility regimes typically result in higher allocations to riskier assets such as equities. In contrast, high-volatility regimes tend to favor defensive assets such as government bonds or gold.

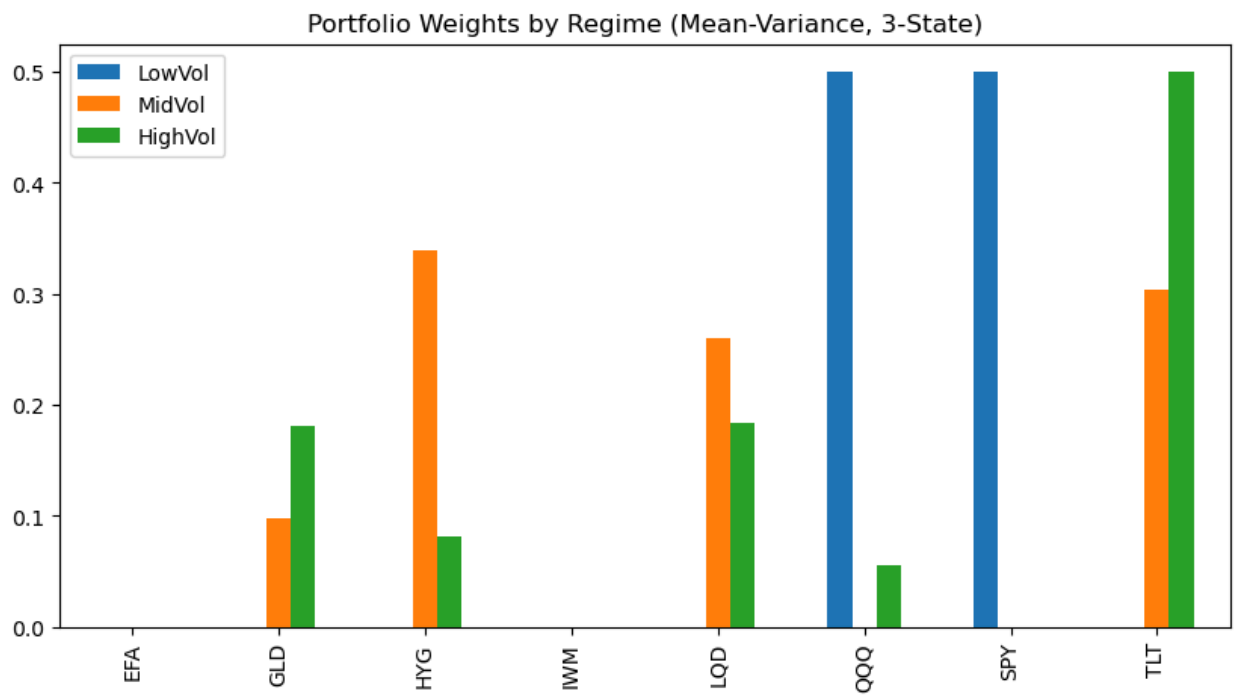

Figure 2: Regime-specific optimal portfolio allocation across assets

To illustrate the differences in portfolio composition across regimes, Figure 2 presents the optimal asset allocations for each regime using the three-state HMM model and mean-variance optimization. The figure shows that the low-volatility regime tends to allocate more heavily toward equities, reflecting favorable market conditions. In contrast, the high-volatility regime shifts allocation toward defensive assets, such as long-term Treasury bonds and gold. Similar allocation compositions also appear in the two-state model. In the three-state model, the medium-volatility regime is observed to allocate little or no weight to equity assets, such as SPY and QQQ. This reflects the regime's relatively unfavorable risk-return profile for

equities, where expected returns are low while volatility remains elevated. As a result, the optimizer shifts toward more defensive assets, producing a more conservative allocation during periods of moderate uncertainty. Notably, some assets receive zero weight across any regime, reflecting their relatively inferior risk-return tradeoff or redundancy with other assets.

Two approaches are used to translate regime information into daily portfolio allocations: a hard regime-switching strategy and a probability-weighted strategy.

Hard Regime-Switching Strategy

In the hard regime-switching approach, the portfolio is fully allocated to the regime-specific portfolio corresponding to the most likely regime at each point in time. Formally, let $\hat{S}_{t-1}$ denote the most likely regime at time $t - 1$. The portfolio allocation at time t is then given by:

$$w_t = w_{\hat{S}_{t-1}}$$

This approach results in discrete changes in portfolio allocation whenever the inferred regime changes. While straightforward to implement, it can be sensitive to misclassification, particularly when the model assigns similar probabilities to multiple regimes.

Probability-Weighted Strategy

To address the limitations of hard-switching, a probability-weighted portfolio construction method is implemented. Instead of selecting a single regime, this method constructs a portfolio as a weighted combination of all regime-specific portfolios, using the posterior probabilities produced by the HMM.

The portfolio weights at time t are given by:

$$w_t = \sum_{k=1}^{K} p_{k,t-1} \cdot w_k$$

where $p_{k,t-1} = P(S_{t-1} = k \mid data)$ represents the probability that the market is in regime k at time $t - 1$, and w_k is the portfolio optimized for regime k.

This formulation ensures that the portfolio reflects the model's degree of confidence in each regime. For example, if the model assigns a high probability to a single regime, the resulting portfolio will closely resemble that regime's optimal allocation. Conversely, if the model is uncertain and assigns similar probabilities across multiple regimes, the portfolio will represent a more balanced combination of allocations. An important feature of this approach is that all probabilities are lagged by one period, ensuring that only information available at time $t - 1$ is used to determine the portfolio at time t. This prevents look-ahead bias and maintains the integrity of the backtesting framework.

The probability weighted strategy offers several advantages over hard-switching. First, it produces smoother transitions between allocations, reducing abrupt changes in portfolio composition. Second, it mitigates the impact of regime misclassification by incorporating uncertainty directly into the allocation process. As a result, this approach is expected to deliver more stable and robust performance in practice.

3.5 Backtesting Framework

To evaluate the performance of the regime-based portfolio strategies, an out-of-sample backtesting framework is implemented. The objective is to simulate how the strategies would have performed in real time, using only information available at each point in time. This approach ensures that the results provide a realistic assessment of performance and are not affected by look-ahead bias.

The backtesting procedure is conducted using daily return data over the out-of-sample period from January 2019 to December 2025. At each time step, portfolio returns are calculated using the asset weights determined by the regime identification process and portfolio construction methodology described in previous sections. A key component of the framework is the use of lagged regime signals. Specifically, regime classifications and posterior probabilities obtained from the HMM are shifted by one period, so that portfolio weights at time t are determined using information available at time $t - 1$. This ensures that the model does not incorporate future information when making allocation decisions and maintains the integrity of the backtest.

Given the portfolio weights w_t and the vector of asset returns r_t, the portfolio return at time t is calculated as:

$$R_t = w_t^T r_t$$

where R_t denotes the portfolio return at time t. Portfolio returns are then aggregated over time to evaluate cumulative performance.

To provide a meaningful benchmark for comparison, two alternative portfolio strategies are constructed. The first is an equal-weight portfolio, in which capital is allocated uniformly across all assets at each point in time. The second is a static mean-variance portfolio, constructed using the full in-sample dataset without conditioning on market regimes. Both benchmark strategies are validated over the same out-of-sample period and use the same return data, ensuring a fair comparison. It is important to note that benchmark portfolios do not require lagging, as their allocations do not depend on time-varying signals. However, all portfolio returns are aligned to the same time index to ensure that performance comparisons are made over identical periods.

Performance is evaluated using a set of standard metrics commonly used in portfolio analysis. These include average daily return, daily volatility, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, and total cumulative return. Annualized return and volatility are calculated using standard scaling factors, while the Sharpe ratio is defined as the ratio of excess return to volatility. In addition to performance measurement, cumulative returns are calculated to visualize the evolution of portfolio value over time. This allows for a direct comparison of growth trajectories across different strategies and provides insights into the timing and magnitude of performance differences.

Overall, this backtesting framework ensures a consistent and unbiased evaluation of all strategies. By combining out-of-sample testing, lagged signals, and appropriate benchmark comparisons, the framework provides a robust basis for assessing the effectiveness of regime-based portfolio optimization.

4. Validation

This section evaluates the performance of the regime-based portfolio strategies using the out-of-sample backtesting framework described in Section 3.5. The objective is to assess whether incorporating regime information leads to improved portfolio performance relative to traditional benchmark strategies. All strategies are evaluated over the same out-of-sample period, ensuring a fair and consistent comparison.

Table 1 summarizes the performance of the different portfolio strategies, including regime-switching and probability-weighted approaches for both two-state and three-state models, as well as benchmark portfolios. The reported metrics include average daily return, volatility, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, and total cumulative return.

Table 1: Out-of-sample performance comparison of portfolio strategies

Portfolio	Mean daily	Vol daily	Annualized Return approx	Annualized Vol approx	Sharpe approx	Max Drawdown	Total return
Regime-switching 2-state	0.0008	0.0083	0.1927	0.1318	1.4628	-0.2899	2.8362
Prob-weighted 2-state	0.0008	0.0078	0.1960	0.1233	1.5892	-0.2593	2.9247
Regime-switching 3-state	0.0006	0.0069	0.1458	0.1091	1.3366	-0.2549	1.7654
Prob-weighted 3-state	0.0007	0.0064	0.1639	0.1018	1.6099	-0.2532	2.1370
Equal Weight	0.0004	0.0079	0.1004	0.1249	0.8033	-0.2500	1.0139
Static Portfolio	0.0003	0.0065	0.0689	0.1027	0.6707	-0.2753	0.6169

Performance Comparison

The results are broadly consistent with the economic intuition developed in Section 3, where different regimes correspond to distinct risk-return environments and portfolio allocations. Table 1 shows that regime-based strategies significantly outperform the benchmark portfolios. Both the two-state and three-state models achieve substantially higher returns and Sharpe ratios compared to the equal-weight and static mean-variance portfolios. This indicates that incorporating regime information provides meaningful improvements in both absolute and risk-adjusted performance.

Among all strategies, the probability-weighted approaches consistently outperform their corresponding hard regime-switching counterparts. This improvement can be attributed to the smoother transitions between regimes and the reduced sensitivity to classification errors. By incorporating the full distribution of regime probabilities, the probability-weighted strategy produces more stable and robust portfolio allocations. Comparing the two-state and three-state models reveals an important tradeoff. The two-state

probability-weighted model achieves the highest annualized return, reflecting a more aggressive allocation strategy during favorable market conditions. In contrast, the three-state probability-weighted model delivers the highest Sharpe ratio and lowest volatility, indicating superior risk-adjusted performance. This suggests that the inclusion of an intermediate regime allows for more gradual adjustments in portfolio allocation, leading to improved risk control.

The benchmark portfolios perform significantly worse across annualized return and Sharpe ratio metrics. The equal-weight portfolio, while diversified, does not adapt to changing market conditions and therefore achieves a lower return and Sharpe ratio. It does, however, achieve the lowest maximum drawdown across all strategies and a comparable annualized volatility to the two-state strategies. The static mean-variance portfolio underperforms the other strategies in nearly every metric except for annualized volatility, where it achieves the second lowest.

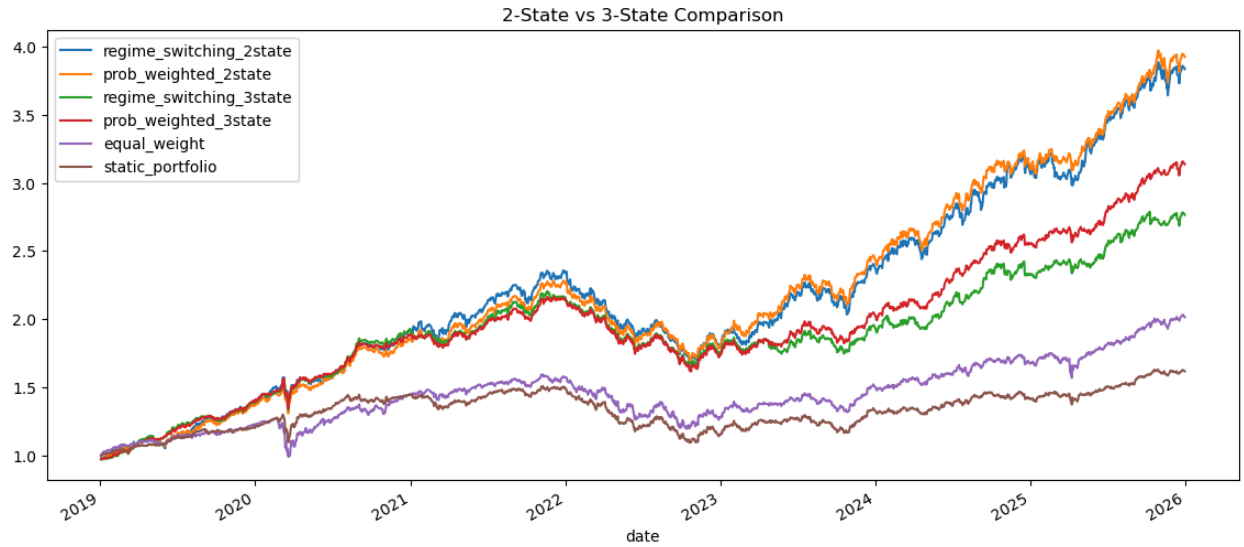

Figure 3: Cumulative returns of portfolio strategies over out-of-sample period

To further illustrate the results, Figure 3 shows the exponentiated cumulative log returns over the out-of-sample period, which represent the compounded growth of a $1 investment over time. This differs from the total return values reported in Table 1, which are based on the sum of simple returns and therefore do not fully capture compounding effects. As shown in the figure, the probability-weighted two-state strategy achieves the highest cumulative growth, with the hard regime-switching two-state strategy close behind. The three-state strategies underperform the two-state strategies from a cumulative growth perspective, but still strongly outperform the benchmark strategies.

Several important insights emerge from the results. First, incorporating regime information into portfolio construction does, in fact, lead to significant improvements in performance. Second, probability-weighted strategies outperform hard regime-switching by accounting for uncertainty in regime classification. Third, increasing model complexity from two to three states improves risk-adjusted performance, but at the cost of lower absolute returns. Finally, the results remain robust when extending the evaluation period, indicating that the model generalizes well across different market conditions.

5. Conclusions

This project examined the use of Hidden Markov Models to identify market regimes and incorporate this information into portfolio optimization. By modeling financial markets as a sequence of latent states characterized by different return and volatility dynamics, the study developed a framework for dynamically adjusting portfolio allocations in response to changing market conditions.

The results demonstrate that incorporating regime information into the portfolio construction process leads to meaningful improvements in performance relative to traditional static strategies. In particular, regime-based approaches are able to adapt to periods of market stress and stability by shifting allocations toward more appropriate asset classes. This dynamic adjustment allows the strategies to better manage risk while maintaining strong return potential. Among the approaches considered, probability-weighted portfolio construction proved to be more effective than hard regime switching. By incorporating the full distribution of regime probabilities, this method produces smoother transitions between allocations and reduces the impact of regime misclassification. This leads to more stable and robust portfolio performance across different market environments. The comparison between two-state and three-state models highlights a tradeoff between simplicity and flexibility. While the two-state model captures broad market conditions effectively, the three-state model provides a more nuanced representation by introducing an intermediate regime. This additional flexibility improves risk management, although it may come at the cost of slightly lower return potential.

Overall, the findings suggest that regime-aware portfolio optimization provides a valuable enhancement to traditional investment strategies. By combining statistical modeling with dynamic asset allocation, the approach offers a practical framework for improving both return and risk characteristics in real-world portfolio management. Several potential extensions could further improve the framework. These include incorporating transaction costs and taxes, expanding the asset universe, and exploring alternative regime detection methods or machine learning approaches. Additionally, allowing for time-varying model parameters or incorporating macroeconomic variables could enhance the model's ability to capture evolving market conditions. These findings highlight the potential for regime-aware models to play an important role in adaptive portfolio management in increasingly dynamic financial markets.

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Appendix

This appendix provides additional details regarding the data, software, and implementation used in this project.

A.1 Data Description

The dataset used in this project consists of daily adjusted closing prices for a set of exchange-traded funds (ETFs) representing different asset classes. The selected ETFs include:

- **SPY** – S&P 500 ETF (U.S. equities)
- **QQQ** – Nasdaq-100 ETF (growth equities)
- **IWM** – Russell 2000 ETF (small-cap equities)
- **EFA** – MSCI EAFE ETF (international equities)
- **TLT** – Long-term U.S. Treasury bonds
- **LQD** – Investment-grade corporate bonds
- **HYG** – High-yield corporate bonds
- **GLD** – Gold ETF

The data spans the period from **January 2007 to December 2025** and was obtained using the Yahoo Finance API via the **yfinance** Python library. Daily returns were computed from adjusted closing prices and used as inputs to the Hidden Markov Model and portfolio optimization procedures.

A.2 Software and Libraries

All computations and analysis were conducted using Python. The primary libraries used in this project include:

- **NumPy** – numerical computations
- **Pandas** – data manipulation and time series handling
- **Matplotlib** – data visualization
- **Scikit-learn** – preprocessing (e.g., StandardScaler)
- **hmmlearn** – Hidden Markov Model implementation
- **CVXPY** – convex optimization for portfolio construction
- **yfinance** – data retrieval

The analysis was performed in a Jupyter Notebook environment, with modular code organized into utility functions for clarity and reusability.

A.3 Portfolio Optimization Implementation

Portfolio weights for each regime were obtained by solving a constrained mean-variance optimization problem using the CVXPY library. The optimization enforces:

- Full investment constraint: $\Sigma w_i = 1$

- Long-only constraint: $w_i \geq 0$
- Optional maximum weight constraint to prevent concentration

The optimization problem is solved separately for each regime using regime-specific estimates of expected returns and covariance matrices.

A.4 Backtesting Implementation

The backtesting framework was implemented using daily return data over the out-of-sample period. Portfolio weights were computed using lagged regime information to avoid look-ahead bias. Portfolio returns were calculated as the weighted sum of asset returns at each time step.

Cumulative returns were computed using compounded returns, representing the growth of a $1 investment over time. Performance metrics, including annualized return, volatility, Sharpe ratio, and maximum drawdown, were calculated to evaluate strategy performance.

A.5 Additional Notes

- All results presented in the report are reproducible using the provided code
- Minor numerical differences may occur due to randomness in model initialization, particularly in the Hidden Markov Model estimation
- No external datasets beyond those obtained from Yahoo Finance were used

A.6 Code Availability

The full implementation code is provided as a separate submission and is organized into a main notebook and supporting utility modules.